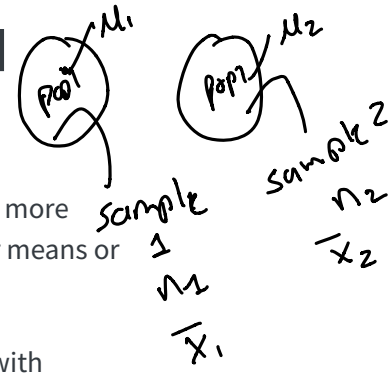
11 Two-sample comparisons and Hypothesis testing



A common scenario in scientific research involves comparison of two or more samples from different populations and it is of interest to compare their means or standard deviations.

- Consider two indpendent random samples of n_1 and n_2 observations with sample means $\bar{x}_1$ and $\bar{x}_2$ and sample standard deviations s_1 and s_2 respectively. Then the standard error of $\bar{X}_1 - \bar{X}_2$ is:

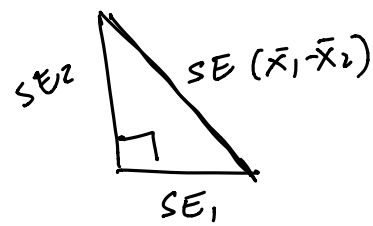
interest: $\mu_1 - \mu_2$

Unpooled standard error (Welch)

$SE^2 = s_1^2 + s_2^2$

$c^2 = a^2 + b^2$

$$\mathrm{SE}\left(\bar{X}_1 - \bar{X}_2\right) = \sqrt{\underbrace{\frac{s_1^2}{n_1}}_{SE_1^2} + \underbrace{\frac{s_2^2}{n_2}}_{SE_2^2}}$$



If you believe that the two populations have the same standard deviations ($\sigma_1 = \sigma_2$) then you can use the pooled SE.

% Pooled standard error for difference in two sample means (equal variances):

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\mathrm{SE}_{\text{pooled}}\left(\bar{X}_1 - \bar{X}_2\right) = s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Example: A new fertilizer

A study tests whether a new fertilizer increases plant growth. After 4 weeks, researchers measure plant height (cm). - Control: $n_C = 20, \bar{x}_C = 15.2, s_C = 2.8$ - Treatment: $n_T = 18, \bar{x}_T = 17.0, s_T = 3.1$

Difference in sample means:

$$\hat{D} = \bar{x}_T - \bar{x}_C = 17.0 - 15.2 = 1.8 \text{ cm}.$$

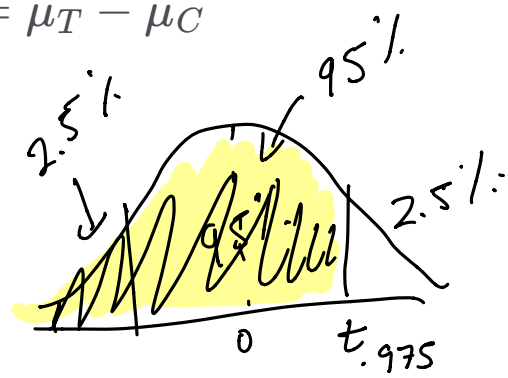
Unpooled standard error:

$$\mathrm{SE} = \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}} = \sqrt{\frac{3.1^2}{18} + \frac{2.8^2}{20}} \approx 0.962.$$

We got a point estimate $\bar{x}_1 - \bar{x}_2$ and standard errors. Let's build confidence intervals for the difference in means!

11.0.1 95% confidence interval for $D = \mu_T - \mu_C$

$$\hat{D} \pm t_{0.975,\,\mathrm{df}}\,\mathrm{SE}.$$



Degrees of freedom:

$$\mathrm{df} = \frac{\left(\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}\right)^2}{\frac{\left(\frac{s_T^2}{n_T}\right)^2}{n_T - 1} + \frac{\left(\frac{s_C^2}{n_C}\right)^2}{n_C - 1}} \approx 34.$$

pnorm → pt

x	P(X≤x)
-2	.025
-1	.16
0	.5
1	
2	

qt, qnorm

P(X≤x)	x
.975	2
.5	0
.025	-2

Using $t_{0.975,34} \approx 2.03$:

$P(T \le t_{.975}) = .975$ 95.7 percentile .975 quantile

$P(T \le 2.03) = .975$

```
qt(.975,34)
```

```
[1] 2.032245
```

1.8 ± t SE

$(\bar{X}_1 - \bar{X}_2) \pm (2.03)(.962)$

$\bar{D}$

```
1.8 - 2.03*(0.962)
```

```
[1] -0.15286
```

```
1.8 + 2.03*(0.962)
```

```
[1] 3.75286
```

95% CI (−.15, 3.75)

$$1.8 \pm 2.03(0.962) = 1.8 \pm 1.95,$$

so the 95% CI is:

$$(-0.15,\ 3.75)\ \mathrm{cm}.$$

What if we believe we got samples from the same population?

11.1 Hypothesis testing: randomization test

The hypothesis is that the two samples actually came from the same population. The labels we applied – “treatment” vs “control” – were arbitrary and could thus be arbitrarily reassigned without problems.

> **Example: Changes in mice diet**
>
> In (Sörhede Winzell and Ahrén 2004), researchers were interested in determining whether following a given diet makes mice heavier after several weeks. The data were generated by obtaining 24 mice and randomly assigning them to either a chow or a high-fat (hf) diet. After several weeks, the scientists weighed each mouse and recorded the results.

Results can be retrieved from R as follows:

```
dir <- "https://raw.githubusercontent.com/genomicsclass/dagdata/
filename <- "femaleMiceWeights.csv"
url <- paste0(dir, filename)
dat <- read.csv(url)
head(dat)
```

```
  Diet Bodyweight
1 chow      21.51
2 chow      28.14
3 chow      24.04
4 chow      23.45
5 chow      23.68
6 chow      19.79
```

```
head(dat)
```

```
  Diet Bodyweight
1 chow      21.51
2 chow      28.14
3 chow      24.04
4 chow      23.45
5 chow      23.68
6 chow      19.79
```

```
## Note that head only shows the first 6/24 observations in the
## Lightest mouse:
dat[which.min(dat$Bodyweight), ]
```

```
  Diet Bodyweight
6 chow      19.79
```

```
## Heaviest mouse
dat[which.max(dat$Bodyweight), ]
```

```
   Diet Bodyweight
21   hf      34.02
```

```
## Use 'View(dat)' in R to see the entire dataset.
```

Notice that there is variability in body weight across the mice. For instance, mouse 6 was the lightest (19.79 grams) and was on the chow diet, while mouse 21 was the heaviest (34.02 grams) and was on the high-fat (HF) diet. However, these are only individual observations. To better understand patterns in the data, we can calculate the *average* weight within each group.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
control <- filter(dat, Diet=="chow") %>% select(Bodyweight) %>%
treatment <- filter(dat, Diet=="hf") %>% select(Bodyweight) %>%
print(mean(control))
```

unlist
unlist

```
[1] 23.81333
```

```r
print(mean(treatment))
```

```
[1] 26.83417
```

```r
obsdiff <- mean(treatment) - mean(control)
print(obsdiff)
```

```
[1] 3.020833
```

In this experiment, we observe that, on average, mice on the high-fat diet are about 10% heavier than those on the chow diet. However, this result is based on a sample drawn from a larger population of mice. If the experiment were repeated with a new sample of 24 mice, randomly assigned to each diet, we would likely obtain a different average. In fact, each repetition of the experiment would yield a different value. ~~This type of quantity is known as a random variable.~~

To illustrate this, suppose we had access to the weights of the entire population of control female mice. In practice, this is rarely the case, as we typically observe only samples and rely on statistical methods to make inferences about population characteristics. For illustrative purposes, assume the following dataset represents the full population of female mice. We can then draw multiple random samples of 12 mice from this population and calculate the average for each sample.

```r
## Download the dataset directly from: https://raw.githubuserco
population <- read.csv("data/femaleControlsPopulation.csv")
## Use unlist to turn it into a numeric vector
population <- unlist(population)
# Mean of the 'total' population
mean(population)
```

```
[1] 23.89338
```

Hypothesis: $\mu_T = \mu_C$

H_0: $D = \mu_T - \mu_C = 0$

```r
# Sample 12 mice three times to see if the average changes.
control <- sample(population, 12)
mean(control)
```

```
[1] 24.25083
```

```r
control <- sample(population, 12)
mean(control)
```

```
[1] 23.175
```

```r
control <- sample(population, 12)
mean(control)
```

```
[1] 24.50917
```

See how the average changes across samples, and that each differs from the average of the 'total' population. Therefore, repeating this experiment is useful, as it provides additional information about the distribution of this random variable.

11.2 Null Hyphotesis

In the previous section, we saw that the true characteristics of a population are rarely known. Instead, we use statistical methods to infer or approximate these characteristics based on samples drawn from that population. In the case of the mice experiment, scientists seek to determine whether applying a different diet leads to increased body weight. Based on their analysis, they conclude the following (Sörhede Winzell and Ahrén 2004):

'Body weight was higher in mice fed the high-fat diet already after the first week, due to higher dietary intake in combination with lower metabolic efficiency.'

This conclusion is based on comparing the average body weight between the control and treatment groups. However, we have also seen that the average is a random variable—it changes each time we draw a different sample from the population. This raises an important question: should we fully trust the scientists' conclusion? How do we know that the observed difference is truly due to the diet? What would happen if all 24 mice were given the same diet—would we still observe a large difference?

Statistics addresses this scenario through the concept of the **null hypothesis**. It is called **"null"** because it represents the possibility that there is no real difference between groups.

We can simulate this idea in by using the full dataset of control mice, which we previously treated as the 'population.' Assuming that diet has no effect, we can repeatedly draw random samples of 24 control mice, assign them (artificially) to two groups of 12, and calculate the difference in average weight between these

groups. This allows us to observe how much variation can arise purely by chance and test the hypothesis:

Null hypothesis: $H_0 : \mu_1 = \mu_2$ (Diet has no effect)

Alternative hypothesis: $H_A : \mu_1 \neq \mu_2$

```
## 12 control mice
control <- sample(population, 12)
## Another 12 control mice that we act as if they were not
treatment <- sample(population, 12)
print(mean(treatment) - mean(control))
```

```
[1] 1.828333
```

```
## Now, let's do it 10,000 times (using a 'for-loop')
n <- 10000
null <- vector("numeric", n)
for (i in 1:n){
  control <- sample(population, 12)
  treatment <- sample(population, 12)
  null[i] <- mean(treatment) - mean(control)
}
```

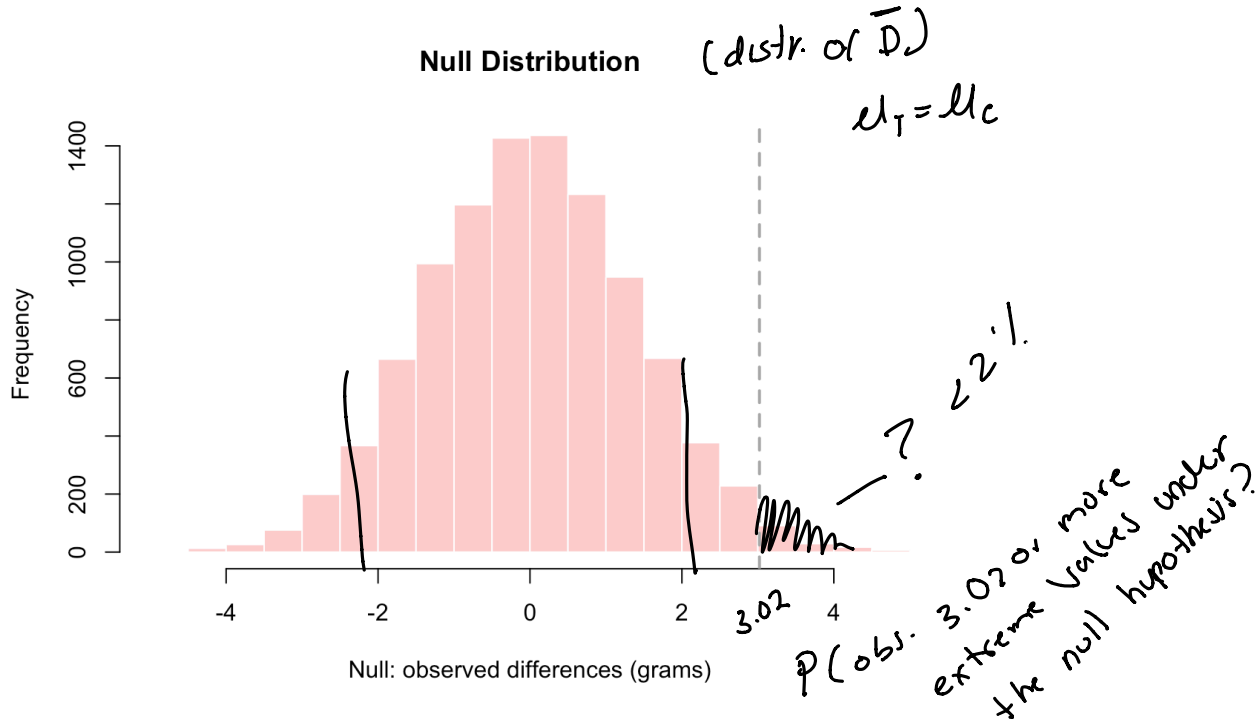
The values under the null represent what we call the null distribution. We can now compare the mean of the null distribution with the difference observed by the scientists in the original experiment (obsdiff = 3.020833). For example, what percentage of the 10,000 sample values in this distribution are greater than 3.020833?

```
mean(null >= obsdiff) * 100
```

```
[1] 1.4
```

```
hist(null,
     mai = "Null Distribution",
     xlab = "Null: observed differences (grams)",
     col = "#FFCCCC",
     border = "white")

# Vertical dashed line at observed value
abline(v = 3.020833,
       col = "darkgrey",
       lwd = 2,
       lty = 2)
```



That is, based on the 10,000 simulations assuming no diet effect, we observe a difference as large as the one in the original experiment only 1.3% of the time. In statistics, this is what is known as a *p-value*.

→ significance = 5%

p-value < 5%

Definition: The P-value for a hypothesis test is the probability, computed under the condition that the null hypothesis is true, of the **test statistic** being at least as extreme as the value of the test statistic that was actually obtained.

Corollaries to definition:

- The P-value is a measure of compatibility between the data and H_0 and thus measures H_A.
- A large P-value (close to 1) indicates the data are compatible with H_0.
- A small P-value (close to 0) indicates the data are incompatible with H_0.

In the previous example we were able to use simulation to estimate the p-value. In many situations, we will not be able to do this. An alternative is the t-test with test statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE_{(\bar{x}_1 - \bar{x}_2)}}$$

```
print(mean(control))
```

```
[1] 24.94833
```

```
print(mean(treatment))
```

```
[1] 23.0675
```

```
obsdiff <- mean(treatment) - mean(control)
print(obsdiff)
```

```
[1] -1.880833
```

```
part1<-(sd(control)^2)/length(control)
part2<-(sd(treatment)^2)/length(treatment)
se<-sqrt(part1+part2)
tval<-obsdiff/se
tval
```

```
[1] -1.520291
```

```
df<-((part1+part2)^2)/(part1^2/(length(control)-1)+part2^2/(leng
df
```

```
[1] 21.7683
```

Let's compute the p-value

```
2*pt(tval,df)
```

```
[1] 0.1428309
```

We can compare our calculations:

```
t.test(control, treatment)
```

```

    Welch Two Sample t-test

data:  control and treatment
t = 1.5203, df = 21.768, p-value = 0.1428
alternative hypothesis: true difference in means is not equal
to 0
95 percent confidence interval:
 -0.686450  4.448117
sample estimates:
mean of x mean of y
 24.94833  23.06750
```

The data is compatible with the hypothesis. We do not reject the null hypothesis of equality in means.

Sörhede Winzell, Maria, and Bo Ahrén. 2004. "The High-Fat Diet-Fed Mouse: A Model for Studying Mechanisms and Treatment of Impaired Glucose Tolerance and Type 2 Diabetes." *Diabetes* 53 (Suppl 3): S215–19. https://doi.org/10.2337/diabetes.53.suppl_3.S215.

n_T = treatment group sample size

$n_C = n_T$

$\bar{X}_T$

$\bar{X}_C$

- σ_T
- σ_C

Goal: Construct a 95% CI for

$$D = \mu_T - \mu_C$$

estimate of D: $\hat{D} = \bar{X}_T - \bar{X}_C$

T	C	D
X_1	Y_1	$X_1 - Y_1$
X_2	Y_2	$X_2 - Y_2$
$\vdots$		

$$x_n \quad y_n \qquad x_3 - y_3$$

$$\bar{X}_T - \bar{X}_C \qquad \bar{D}$$

$$SE(\bar{D}) = SE(\overbrace{\bar{X}_T - \bar{X}_C}^{\bar{D}})$$

$$= \frac{SD_D}{\sqrt{n}} = \frac{\sqrt{\sigma_T^2 + \sigma_C^2}}{\sqrt{n}}$$

- $$Var(\bar{X}_T - \bar{X}_C) = Var(\bar{X}_T) + Var(\bar{X}_C)$$

$$= \frac{\sigma_T^2}{n} + \frac{\sigma_C^2}{n}$$

Reminder

$$Var(X_1 + X_2) = Var(X_1) + Var(X_2)$$

$$Var(\underline{X_1} - \underline{X_2}) = Var(X_1) + Var(X_2)$$

$$Var(-X_1) = Var(X_1)$$

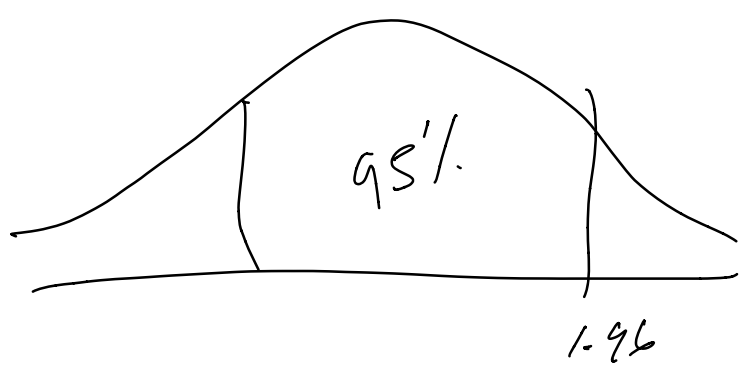
$$var(cX) = c^2 var(X)$$

$$var(X_1 \pm (-1)(X_2))$$

$$= var(X_1) + var((-1)X_2)$$

95% CI

$$\bar{D} \pm 1.96\, SE_D$$

true SE

$$\sqrt{\frac{\sigma_T^2 + \sigma_C^2}{n}}$$